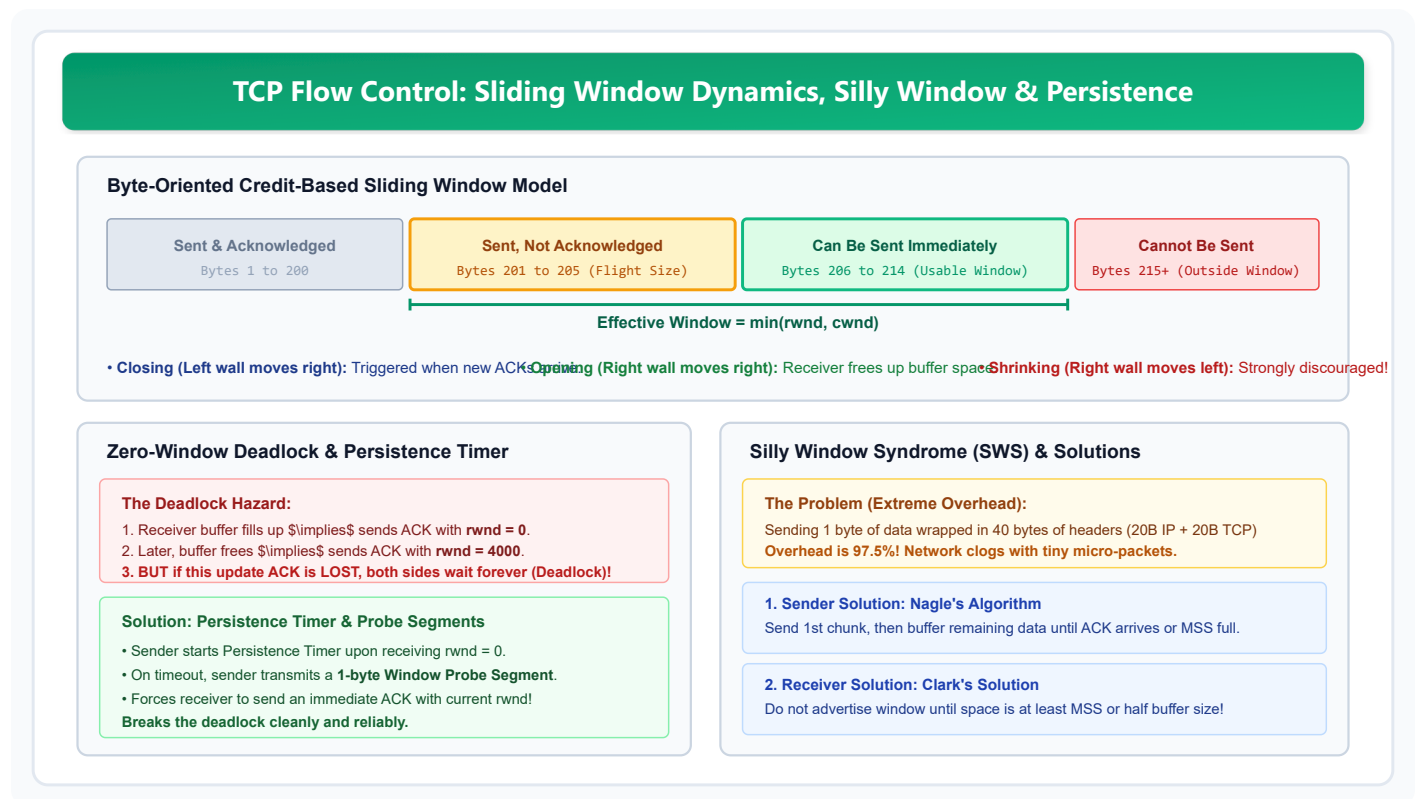


Module 06: TCP Flow Control and Sliding Window Mechanics

Unit 4: Transport Layer

Flow Control & Buffer Management

Credit-Based Byte-Oriented Sliding Window, Window Opening/Closing/Shrinking Dynamics, Zero-Window Deadlock Prevention via Persistence Timer, and Silly Window Syndrome.



1. Sliding Window Wall Dynamics

Window Size Equation: $\text{Effective Window} = \min(\text{rwnd}, \text{cwnd})$

Closing: Left wall moves right upon receiving ACKs for sent bytes.

Opening: Right wall moves right as receiver application drains buffer.

Shrinking: Right wall moves left (Forbidden / strongly discouraged by RFCs).

2. Silly Window Syndrome Mitigations

Technique	Side Applied	Rule Enforced
Nagle's Algorithm	Sender Side	Buffer small chunks until previous ACK returns or MSS accumulated
Clark's Solution	Receiver Side	Do not advertise window until space $\geq$ MSS or $\geq$ half buffer space